

RESEARCH ARTICLE

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Key Points:

- New procedure for mapping electrodynamics with SuperDARN and AMPERE data
- Various conductivity models evaluated quantitatively
- Using two data sets gives demonstrably better results than using just one

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Mapping high-latitude ionospheric electro-dynamics with SuperDARN and AMPERE

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Abstract An assimilative procedure for mapping high-latitude ionospheric electro-dynamics is developed for use with plasma drift observations from the Super Dual Auroral Radar Network (SuperDARN) and magnetic perturbation observations from the Active Magnetosphere and Planetary Electrodynamics Response Experiment (AMPERE). This procedure incorporates the observations and their errors, as well as two background models and their error covariances (estimated through empirical orthogonal function analysis) to infer complete distributions of electrostatic potential and vector magnetic potential in the high-latitude ionosphere. The assimilative technique also enables objective error analysis of the results. Various methods of specifying height-integrated ionospheric conductivity, which is required by the procedure, are implemented and evaluated quantitatively. The benefits of using both SuperDARN and AMPERE data to solve for both electrostatic and vector magnetic potentials, rather than using the data sets independently or solving for just electrostatic potential, are demonstrated. Specifically, solving for vector magnetic potential improves the specification of field-aligned currents (FACs), and using both data sets together improves the specification of features in regions lacking one type of data (SuperDARN or AMPERE). Additionally, using the data sets together results in a better correspondence between large-scale features in the electrostatic potential distribution and those in the FAC distribution, as compared to using SuperDARN data alone to infer electrostatic potential and AMPERE data alone to infer FACs. Finally, the estimated uncertainty in the results decreases by typically ~20% when both data sets rather than just one are included.

1. Introduction

Electric fields and currents in the Earth's high-latitude ionosphere, which are highly variable in space and time, are a significant source of energy and momentum to the upper atmosphere. The behavior of these fields also provides a window into processes in the magnetosphere and interactions between the magnetosphere and the solar wind, due to the current flow and transmission of electric fields along highly conducting geomagnetic field lines. As a result, various efforts have been made to estimate complete, instantaneous distributions of electrodynamic parameters across the high-latitude regions, based on various data sets used individually or in conjunction with one another.

Some such procedures for "mapping" electrodynamic parameters have made use of individual data sets with large spatial coverage, such as plasma drift observations from the Super Dual Auroral Radar Network (SuperDARN) [Ruohoniemi and Baker, 1998; Cousins et al., 2013a] or magnetic perturbation observations from the Active Magnetosphere and Planetary Electrodynamics Response Experiment (AMPERE) [Anderson et al., 2014]; these procedures that use only one data type typically only estimate one electrodynamic variable (e.g., electric fields or field-aligned currents (FACs)). The Assimilative Mapping of Ionospheric Electrodynamics (AMIE) procedure developed by Richmond and Kamide [1988], on the other hand, incorporates various data types with various spatial coverages and can estimate various electrodynamic parameters (e.g., electric fields and FACs) simultaneously.

In this study, we follow the approach of Richmond and Kamide [1988] to develop an assimilative mapping procedure, but we develop it specifically for SuperDARN and AMPERE data combined. These data sets are both routinely available with large spatial coverage at a relatively high temporal cadence, and together they can constrain both the fundamental variables of electrostatic potential and vector magnetic potential in the high-latitude ionosphere, from which electric fields, magnetic perturbations, and FACs can be calculated.

One significant challenge, however, is introduced when incorporating both electric field and magnetic perturbation data—knowledge of the height-integrated ionospheric conductivity (conductance) is required. This parameter is highly variable in space and time, and observations are very sparse, such that accurately specifying the instantaneous conductance distribution presents a significant challenge. In this study, we implement and quantitatively evaluate several methods for specifying conductance.

Despite large uncertainties in specifying conductance, incorporating the plasma drift and magnetic perturbation data sets together in a mapping procedure can provide multiple significant benefits over working with the data sets independently: the data coverages can complement each other, filling in each others gaps, and the different electrodynamic variables (e.g., electric fields and magnetic perturbations) can be estimated simultaneously and consistently.

The assimilative approach taken in this study provides a framework for optimally combining the different observations, as well as background statistical models, taking into account the error properties of the various sources of information. In particular, the procedure developed in this study incorporates statistical error properties obtained through empirical orthogonal function (EOF) analysis of both SuperDARN data (described by Cousins *et al.* [2013b]) and of AMPERE data (described in the companion paper Cousins *et al.* [2015], referred to hereafter as Paper 1).

The assimilation procedure is described in detail in section 2, the specification of conductance is described and discussed in section 3, the benefits of the approach taken in this study are demonstrated in section 4, and an example application is shown in section 5.

2. Assimilation Procedure

2.1. Data Sources

As described by Anderson *et al.* [2014] and Paper 1, AMPERE data originate from engineering magnetometers onboard spacecraft in the Iridium constellation, which consists of 66 satellites (and additional on-orbit spares) in 6 orbital planes in circular polar orbits at 780 km altitude. This configuration results in 12 cuts in local time that are visited by different spacecraft at a time cadence of 9 mins. In the standard AMPERE mode, magnetometer data are acquired from all spacecraft at a sampling interval of 19.4 s/sample, resulting in a spatial resolution of $\sim 1^\circ$. A high-rate AMPERE mode is also occasionally employed, during which the sampling interval is decreased to 2.16 s/sample, resulting in a $\sim 0.1^\circ$ resolution.

The processing of Iridium magnetometer data by the AMPERE Science Data Center to obtain magnetic perturbation vectors is described by Anderson *et al.* [2014]. For this study, the spacecraft locations and magnetic perturbation vectors are obtained in the original Earth-centered inertial coordinates and are converted into Modified Apex Coordinates referenced to 110 km altitude (as defined by Richmond [1995a]) (referred to hereafter as APEX coordinates), following the method described by Knipp *et al.* [2014]. Only data from locations poleward of $\pm 45^\circ$ latitude and only the two horizontal components (approximately in the magnetic eastward and equatorward directions) of the vector magnetic perturbations are retained for analysis. Finally, as described in Paper 1, the along-track component of the magnetic perturbation vectors is removed due to the increased measurement uncertainties in this component.

As described by Greenwald *et al.* [1995], Chisham *et al.* [2007], and Lester [2013], SuperDARN is an international network of over 30 ground-based high-frequency (HF) radars located in the midlatitude and high-latitude regions of the Northern and Southern Hemispheres. These radars observe the line-of-sight (LOS) component of the $\mathbf{E} \times \mathbf{B}$ drift of plasma in the *F* region ionosphere with a typical spatial resolution of 45 km in the LOS direction and a typical temporal resolution of 1 min. Nominally, the radars operate continuously.

Following the standard SuperDARN procedure described by Ruohoniemi and Baker [1998], the LOS plasma drift data are averaged to a 2 min time resolution and are mapped and averaged onto an equal-area grid in latitude and magnetic local time (MLT) in Altitude Adjusted Corrected Geomagnetic (AACGM) coordinates, which are referenced to the ground [Baker and Wing, 1989]. The grid has a spatial resolution of 1° in latitude (~ 110 km) and an equivalent resolution in longitudinal (~ 110 km). Following Cousins *et al.* [2013b], five 2 min resolution records are concatenated to obtain a 10 min time resolution. This lower time resolution allows for improved SuperDARN data coverage and more closely matches the resampling time resolution of the AMPERE data.

Although the AACGM coordinate system used with SuperDARN data is not equivalent to the APEX coordinate system used with AMPERE data, the two coordinate systems are very similar in the high-latitude ionosphere at ionospheric heights. Any differences in the coordinate systems are expected to be smaller than the uncertainties in geolocating SuperDARN data (~ 100 km) [e.g., *Chisham et al.*, 2008] and thus are considered negligible.

As in Paper 1, this study focuses on the week of 26 November through 2 December 2011. As described in Paper 1, this period contained primarily quiet geomagnetic conditions, with one moderate disturbance, and is representative of typical geomagnetic conditions. AMPERE data are obtained for the complete week, during which period high-rate data are available for 1 day (29 November); the distribution of AMPERE data during this week is shown in Paper 1. SuperDARN data are also available from both hemispheres for every 10 min period in the complete week. In the Northern Hemisphere, which was in winter conditions during the period of interest, excellent SuperDARN data coverage is available (~ 1000 data points in every 10 min period). Fewer SuperDARN observations were made in the Southern Hemisphere (~ 300 data points in every 10 min period), due to the smaller number of radars and a seasonal effect on backscatter occurrence [Ruohoniemi and Greenwald, 1997].

2.2. Optimal Interpolation Technique

To estimate the instantaneous, complete distribution of electrodynamic variables in the high-latitude ionosphere based on any available AMPERE and SuperDARN data, a data assimilation procedure is developed using the optimal interpolation (OI) technique. This procedure closely follows the AMIE procedure developed by *Richmond and Kamide* [1988], the approach of *Matsuo et al.* [2005], and the SuperDARN Assimilative Mapping (SAM) procedure developed by *Cousins et al.* [2013a]. The OI technique combines information from observations and a background model, taking into account the error properties of each source of information, to obtain a solution that is optimal in the least squares sense [e.g., *Kalnay*, 2003]. In this study, as in the works of *Richmond and Kamide* [1988], *Matsuo et al.* [2005], and *Cousins et al.* [2013a], solutions are represented by expansions of basis functions, which will be defined later.

In general terms, the OI technique minimizes the cost function given by

$$J(\alpha) = \frac{1}{2}[\mathbf{d} - \mathbf{H}\alpha]^T \mathbf{R}^{-1}[\mathbf{d} - \mathbf{H}\alpha] + \frac{1}{2}\alpha^T \mathbf{P}_b^{-1} \alpha \quad \text{and} \quad (1)$$

$$\mathbf{d} = \mathbf{y}_{\text{obs}} - \mathbf{y}_{\text{mod}}, \quad (2)$$

where α are the basis function coefficients, $\mathbf{H}$ represents a linear mapping from basis function space to observational space, $\mathbf{y}_{\text{obs}}$ and $\mathbf{y}_{\text{mod}}$ are the values of the observations and the background model, respectively, $\mathbf{R}$ is the observational error covariance matrix, and $\mathbf{P}_b$ is the background model error covariance matrix. The basis function coefficients that minimize equation (1) are referred to as the analysis, α_a , and are given by

$$\alpha_a = \mathbf{K}\mathbf{d} \quad \text{and} \quad (3)$$

$$\mathbf{K} = [\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H} + \mathbf{P}_b^{-1}]^{-1} \mathbf{H}^T \mathbf{R}^{-1}. \quad (4)$$

The error covariance matrix of the analysis, $\mathbf{P}_a$, is then given by

$$\mathbf{P}_a = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_b, \quad (5)$$

where $\mathbf{I}$ is the identity matrix. A more complete description and derivation of the OI equations is given by *Richmond and Kamide* [1988] and *Matsuo et al.* [2005].

The observations used in the OI procedure in this study are AMPERE magnetic perturbations and SuperDARN plasma drift observations, as described in section 2.1. Solutions, which are represented by expansions of basis functions, can be expressed in terms of either electrostatic potential or the radial component of the vector magnetic potential, both of which are scalars. As in Paper 1, we assume for tractability that the two-dimensional magnetic perturbation vectors are purely toroidal and that the relatively small poloidal component produced mainly by the auroral electrojets can be ignored, as it decays fairly rapidly with increasing altitude [e.g., *Langel*, 1974; *Vennerstrom et al.*, 2006], dropping to typically a few tens of nanoteslas at the

780 km altitude of the Iridium spacecraft. With the approximation of vertical magnetic field lines, the relationships between the magnetic perturbation values, $\delta\vec{B}$, the plasma drift values, $\vec{v}$, the electrostatic potential, Φ , and the radial component of the vector magnetic potential, A_r , referred to hereafter simply as magnetic potential, are given by

$$\delta\vec{B} = \nabla \times (\hat{r}A_r), \quad (6)$$

$$\mu_0 J_r = \hat{r} \cdot \nabla \times \delta\vec{B} = -\nabla^2 A_r, \quad (7)$$

$$J_r = \nabla \cdot \vec{I}, \quad (8)$$

$$\vec{I} = \Sigma \cdot \vec{E}, \quad (9)$$

$$\vec{E} = -\nabla\Phi, \quad (10)$$

$$\vec{v} = \frac{\vec{E} \times \vec{B}}{B^2}, \text{ and} \quad (11)$$

$$\nabla^2 A_r / \mu_0 = \Sigma_p \nabla^2 \Phi + \nabla\Phi \cdot \nabla\Sigma_p \pm \hat{r} \cdot (\nabla\Sigma_H \times \nabla\Phi), \quad (12)$$

where μ_0 is the permeability of free space, $\hat{r}$ is a unit radial vector, J_r is the radial current density above the ionosphere (which is assumed to be equivalent to FAC density), $\vec{I}$ is the height-integrated horizontal current density flowing in the ionospheric layer, $\vec{E}$ is the horizontal electric field in the ionosphere, Σ is the conductance in the ionosphere and is a tensor, $\vec{B}$ is the geomagnetic field (given by the International Geomagnetic Reference Field (IGRF)), and Σ_p and Σ_H are the Pedersen and Hall conductances, respectively. The + and – signs in equation (12) are for the Northern and Southern Hemispheres, respectively.

Note that only the horizontal components of the Laplacian, divergence, and gradient operators are included and that equation (12) is obtained by combining equations (7)–(10). In this formulation, any effects of neutral winds are neglected and magnetic field lines are assumed to be radial.

If the ionospheric conductance is known, equations (6)–(12) are linear in terms of either electrostatic potential or magnetic potential. In practice, the ionospheric conductance is unknown, but to maintain linearity, it is treated as a known constant (i.e., it is specified rather than solved for in the OI inversion). Methods of specifying the conductance using empirical models and based on observations are discussed in section 3.

Figure 1 gives a diagram of the primary variables involved in the OI assimilation procedure developed in this study. The options to solve in terms of electrostatic potential or in terms of magnetic potential are shown in Figures 1a and 1b, respectively. The diagram also indicates where knowledge of conductance is required to relate a particular pair of variables. Due to the significant uncertainty involved in specifying conductance, plasma drift observations and electric fields are more directly related to electrostatic potential and magnetic perturbation observations and FACs are more directly related to magnetic potential. Previous studies employing an assimilative procedure with plasma drift or electric field observations and magnetic perturbation observations have worked solely in terms of electrostatic potential [e.g., *Richmond and Kamide*, 1988]. In this study, both electrostatic and magnetic potential will be used, as described in section 4. To account for the significant uncertainty in the conductance distribution, and because the conductance is treated as a constant and cannot have an uncertainty associated with it in the OI procedure, the weight of the AMPERE observations are decreased by a factor of 2 (the errors are increased by $\sqrt{2}$) when solving in terms of electrostatic potential and the SuperDARN observations are similarly dewighted by a factor of 2 when solving in terms of magnetic potential. This factor is estimated empirically, and the exact value is not found to have a significant impact on the results.

As described previously in this section, the OI technique uses information from a background model in addition to the observations. In this study, two different background models are used—one electrostatic potential background model and one magnetic potential background model. When working in terms of magnetic potential, the universal mean magnetic potential distribution derived from AMPERE data as described in Paper

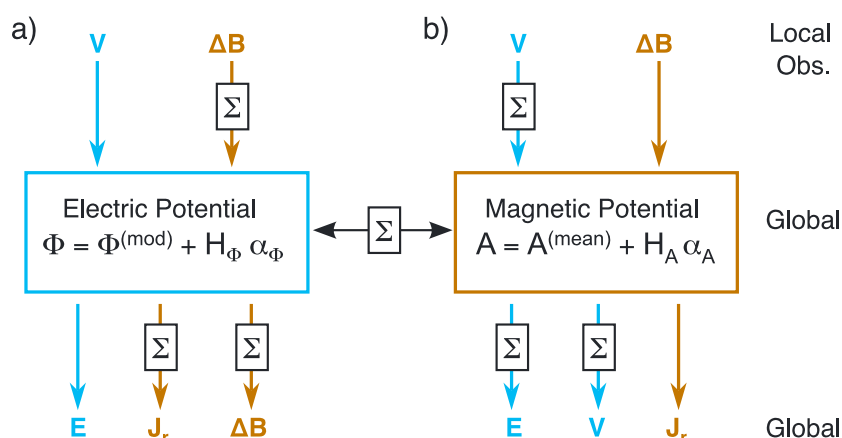


Figure 1. Diagram of variables involved in the OI procedure. Variables at the top represent local observations, from which a global (a) electric or (b) magnetic potential distribution (expressed as an expansion of basis functions, H) can be estimated using the OI technique (solving equations (4) and (3)). From the global potential distributions, global distributions of electrodynamic variables of interest can be calculated, as indicated at the bottom of the figure. Boxed “ Σ ” symbols indicate where knowledge of conductance is required to relate a pair of variables.

1 is used as the background model. When working in terms of electrostatic potential, the empirical convection model developed by *Cousins and Shepherd* [2010] using SuperDARN data (referred to hereafter as the CS10 model) is used. This model, unlike the mean magnetic potential distribution, is parameterized by geophysical parameter values (including solar wind conditions, hemisphere, and dipole tilt—a season-like parameter) and thus varies with time. This is the same background model as is used in the SAM procedure described by *Cousins et al.* [2013a].

Finally, the OI technique relies on error information for both the observations and the background model. The SuperDARN observational error covariance and the CS10 model error covariance used in this study are the same as those used in the SAM procedure described by *Cousins et al.* [2013a]. As described in more detail by *Cousins et al.* [2013a], the observational errors (which encompass both measurement errors and truncation errors due to the use of a finite set of basis functions) are estimated from subresolution variability in the plasma drift measurements and are assumed to be independent. The CS10 model error covariance is defined based on EOF analysis of a ~ 2 year data set of SuperDARN observations. One difference from the errors used by *Cousins et al.* [2013a] is that the observational error variances are multiplied by a factor of 2 for this study. This factor is included because SuperDARN observations have been found to be somewhat inconsistent with similar observations from other instruments [e.g., *Drayton et al.*, 2005], likely due to uncertainties in geolocating the SuperDARN observations and due to the effect of the nonunitary ionospheric index of refraction on the HF radar signal, which is not accounted for in SuperDARN data processing [*Gillies et al.*, 2009, 2011]. Such inconsistencies suggest that a SuperDARN-only variability-based estimate of observational error would likely underestimate the true error.

The AMPERE observational and background model error covariances are estimated using a similar method to that used for SuperDARN errors. AMPERE observational errors are estimated based on small-scale local variability, as described in Paper 1. As with SuperDARN observations, the AMPERE observational errors are assumed to be independent in order to keep the error covariance matrix diagonal and keep the analysis tractable. Also, following *Cousins et al.* [2013a], the observational error variances are scaled by a constant factor at each time step depending on the number of observations available, with the scaling factor given by $\sqrt[4]{L}$, where L is the number of observations. This factor is ~ 4 and ~ 8 for standard and high-rate data, respectively, and accounts for components of observational error not captured by the variability-based estimate (e.g., measurement errors) and helps prevent overweighting of the high-rate AMPERE data, which have a small spatial resolution compared with that of the analysis described in this study and are not likely independent as assumed previously.

The error covariance of the AMPERE-based background model is estimated based on the EOF analysis described in Paper 1. The set of universal EOFs described therein, together with the covariance matrix of their coefficient values, describes the error covariance of the universal mean magnetic potential distribution, which

is used as a background model in this study. Because the EOFs are orthogonal, the EOF coefficient covariance matrix is diagonal. Following *Matsuo et al.* [2005] and *Cousins et al.* [2013a], the diagonal elements of this covariance matrix (the EOF coefficient variances) are approximated as a power law in EOF order v : $V_v = V_1 v^b$, where V_v are the EOF coefficient variances. The parameters of the power law, V_1 and b , are estimated to be 0.002 Tm and -2 , respectively, using the data set of universal EOF coefficients described in Paper 1. It should be noted that only 1 week of data was available for the EOF analysis described in Paper 1, such that the error covariance model implemented in this study may not be universally representative. In the future, the EOF analysis could be simply reperformed using a larger data set to obtain a possibly more universal error covariance model.

In solving the OI equations described previously in this section (equations (1)–(5)), EOFs are used as the basis functions. In this formulation, the background error covariance matrix, $\mathbf{P}_b$, is simply the covariance matrix of the EOF coefficients. This approach minimizes the computational cost of the evaluation, because the EOF coefficient covariance matrix is diagonal and small (30×30 or 10×10 for electrostatic or magnetic potential EOFs, respectively) compared to the error covariance matrix in terms of the original AMIE basis functions of *Richmond and Kamide* [1988] (244×244).

3. Conductance Specification and Evaluation

As described in section 2.2, the mapping procedure developed in this study requires that the ionospheric conductance be completely specified across the high-latitude region. Because the conductance is not, in fact, known completely at any given time, it must be specified approximately; methods of specifying and evaluating the conductance are described in this section.

Conductance in the high-latitude ionosphere has two primary sources, solar irradiance, and auroral particle precipitation. These two contributions to conductivity are specified separately and combined to obtain the total conductance distribution.

The solar-produced conductance is specified using an empirical model parameterized by solar zenith angle and the solar radio flux index, $F_{10.7}$. Various parameterization have been developed based on incoherent scatter radar (ISR) data [see *Brekke and Moen*, 1993, and references therein]. The parametrization used in this study is based on Chatanika ISR data and has been used in the AMIE procedure and is given as

$$\Sigma_p(\chi, F_{10.7}) = \begin{cases} 0.5F_{10.7}^{2/3} \cos(\chi)^{2/3} & \text{for } \chi \leq 65, \\ \Sigma_p(65, F_{10.7}) - 0.22(\chi - 65) & \text{for } 65 < \chi \leq 100, \\ \Sigma_p(100, F_{10.7}) - 0.13(\chi - 100) & \text{for } \chi > 100, \end{cases} \quad (13)$$

$$\Sigma_H(\chi, F_{10.7}) = \begin{cases} 1.8F_{10.7}^{1/2} \cos(\chi) & \text{for } \chi \leq 65, \\ \Sigma_H(65, F_{10.7}) - 0.27(\chi - 65) & \text{for } \chi > 65, \end{cases} \quad (14)$$

where χ is the solar zenith angle in degrees and the conductance values are given in siemens. The values are not allowed to fall below 0.4 S and 0.8 S for Σ_p and Σ_H , respectively. As in the standard AMIE procedure, the conductance values given by equations (13) and (14) are modified slightly to account for an inverse relationship with the geomagnetic field strength [e.g., *Wallis and Budzinski*, 1981; *Richmond*, 1995b].

Empirical models have also been developed to describe the auroral component of the ionospheric conductivity, often based on satellite observations of particle precipitation [e.g., *Fuller-Rowell and Evans*, 1987; *Hardy et al.*, 1987]. Empirical models such as these, however, represent smooth averages and cannot accurately represent any particular instantaneous distribution of auroral-produced conductance, which is highly variable in space and in time. Efforts have been made to estimate the instantaneous auroral-produced conductance based on space-based ultraviolet (UV) images of the auroral region [e.g., *Lummerzheim et al.*, 1991; *Germany et al.*, 1994; *Aksnes et al.*, 2002; *Coumans et al.*, 2004]. Significant uncertainties, however, are involved in this technique [e.g., *Lummerzheim and Liliensten*, 1994; *Germany et al.*, 2001], and the necessary observations are often not available. Due to complicating factors such as these, incorporating conductance observations into the SuperDARN and AMPERE mapping procedure developed in this study is beyond the scope of this paper and is left to future work.

Several options for specifying the auroral component of ionospheric conductivity are implemented in this study: no aurora, aurora from an empirical model, and estimating or adjusting auroral conductivity based

on the AMPERE and SuperDARN observations. The empirical models used in this study are the OVATION Prime [Newell *et al.*, 2009, 2010] and OVATION-SM [Mitchell *et al.*, 2013] models, both based on a large database of Defense Meteorological Satellite Program particle precipitation data but using somewhat different parametrizations. Because OVATION Prime and OVATION-SM are models of auroral precipitation, the resulting conductance is determined using the empirical relationship developed by Robinson *et al.* [1987] to relate particle flux and energy (which are outputs of the OVATION models) to conductance.

Finally, several techniques are tested for estimating or adjusting auroral conductance based on the AMPERE and SuperDARN observations. Electrostatic potential and magnetic potential can be estimated independently using SuperDARN and AMPERE data, respectively, and conductance can be estimated from these two potentials, although the Pedersen conductance is constrained more strongly than is the Hall conductance (see equation (12)). A similar approach has been investigated previously by Green *et al.* [2007]. Significant uncertainties in the estimated conductance are introduced, however, by the sparse and often nonoverlapping coverages of the two data sets and by possible measurement biases in the two data types. For this work, attempts to calculate conductance using this approach resulted in unphysical values of conductance (i.e., negative or extremely large), likely due to the aforementioned uncertainties, and thus, this approach is not used for the mapping procedure developed in this study.

Other techniques to specify or adjust conductivity are based on expected correspondences between features in conductance distributions and features in FAC or electric field distributions. Conductance, for example, is expected to be enhanced in regions of upward FAC [e.g., Blomberg and Marklund, 1988; Marklund *et al.*, 1988], and therefore, it is reasonable to increase conductance above the background solar-produced level in upward regions of a FAC distribution estimated from AMPERE data. The magnitude of the enhancement is related to the FAC density; in this study we use a simple relationship following the form given by Ridley *et al.* [2004]: $\Sigma = \Sigma_0 e^{-A|J_{\parallel}|}$, where Σ_0 and A are constants and $J_{\parallel}$ is FAC density. In this study, the values of Σ_0 and A are fixed at 3 S and $1 \mu\text{A}^{-1}$, respectively. The exact values of the constants in this formula, and even the use of this relationship rather than other relationships between conductance enhancement and FAC density, do not significantly impact the results described below.

Conductance variations may also correspond to features in the electric field distributions. Merkin *et al.* [2005] proposed that conductance can be enhanced in regions of strong electric fields due to anomalous electron heating resulting from the Farley-Buneman instability. We therefore also test increasing the background conductivity based on electric field magnitude as estimated from SuperDARN data, using the dependence given by Merkin *et al.* [2005]. This dependence is expressed as, $\Sigma_{p,H}/\Sigma_{p,H}^0 = \sqrt{E_C/E_{th}}$, where $\Sigma_{p,H}^0$ are the background conductivities, E_C is the convection electric field magnitude, and E_{th} is the instability threshold electric field (a parameter for which various values were tested; a value of 15 mV/m is used in an example below).

To evaluate the various conductance models and adjustments described previously in this section, a quantitative metric is developed based on comparing AMPERE and SuperDARN data. Specifically, the OI equations described in section 2.2 are solved in terms of electrostatic potential using SuperDARN data only, and the estimated electrostatic potential distribution, together with the conductance model of interest, is used to predict magnetic perturbations at AMPERE locations. The SuperDARN-derived magnetic perturbation predictions are compared to the actual AMPERE observations, and the median absolute difference between the two is calculated for all times in the week-long study period. The same test is also carried out in reverse, using magnetic potential estimated from AMPERE data to predict SuperDARN plasma drift observations. Any discrepancies between predictions and observations in these tests can result from inaccuracies in the conductance model, in addition to measurement uncertainties and biases as well as neutral wind effects, which are neglected in this study. Comparing the test results when varying only the conductance model provides one way to quantitatively select the "best" conductance model for the purposes of the procedure developed in this study. It is important to note, however, that the conductance model that minimizes the discrepancies between the AMPERE and SuperDARN data sets may not be the most physically accurate model but rather may be to some extent compensating for measurement biases in the two data sets.

A summary of the test results is given in Table 1, which lists only an illustrative selection of the conductance model variants that have been tested. All the conductance model variants include the same solar-produced conductance, and all except for variants VII and VIII include the diffuse component of aurora from the OVATION Prime model. Variants I and II include conductivity enhancements based on features in the estimated FAC distribution or electric field distribution, respectively, as described above. Variants labeled $\Sigma > x$ indicate the

Table 1. Conductance Model Evaluation^a

Conductance Model	I FAC Adjustment	II E Adjustment	III $\Sigma > 0.4$	IV $\Sigma > 2$	V $\Sigma > 4$	VI $\Sigma > 6$	VII OVATION-SM	VIII No Aurora
$V \rightarrow \delta B$	33.3	38.6	33.2	33.5	34.7	36.7	34.6	34.7
$\delta B \rightarrow V$	501	723	512	175	147	142	146	147

^a Median absolute errors are given for using SuperDARN data to predict AMPERE data ($V \rightarrow \delta B$), in nT, and vice versa ($\delta B \rightarrow V$), in m/s, with estimated uncertainty values of ~ 0.2 nT and ~ 1 m/s, respectively.

background conductance level (of both Σ_p and Σ_H) that the nightside is assumed to take (the original model assumes $\Sigma_p > 0.4$ S and $\Sigma_H > 0.8$ S). Variants VII and VIII both have a background level of 4 S, and VII includes diffuse aurora from the OVATION-SM model.

Comparing variants I and II with variant III, it is clear that adjusting conductance based on features in the FAC or electric field distribution does not significantly improve agreement between the AMPERE and SuperDARN data sets, with electric field-based adjustments actually degrading agreement. We cannot presently explain why these attempts to account for expected approximate physical relations between the conductances and the electric fields and currents does not significantly improve the agreement between the SuperDARN and AMPERE data sets. Perhaps the actual physical relations are much more complex than our attempted parameterizations.

Because the true instantaneous auroral enhancements in conductivity cannot be accurately specified, we test artificially increasing the background conductance everywhere on the nightside to a level similar to long-term average levels for the nightside auroral to subauroral region [Davies and Lester, 1999]. Although no conductance model minimizes prediction errors for both tests, setting the background nightside conductance level to 4 S gives near-optimal error results for both tests. With this background level, little difference is seen between including diffuse aurora from the OVATION Prime model (variant V) or from the OVATION-SM model (variant VII) and including no aurora (variant VIII). Note that including other types of auroral precipitation from the auroral models, in addition to diffuse aurora, has no significant impact on the results.

Based on the results of the conductance model evaluations shown in Table 1, variants V, VII, and VIII give error values that are indistinguishable (after taking into account uncertainty). Given that these three variants cannot be distinguished by a statistical error value, qualitative factors are considered in making a selection. Using a model with some auroral contribution (either OVATION Prime or OVATION-SM) is considered more physical than completely excluding auroral contributions. Finally, Variant V is selected over variant VII because the auroral model used in variant V (OVATION Prime) uses more commonly available inputs than that used in variant VII (OVATION-SM). Therefore, variant V (a background nightside conductance level of 4 S and diffuse aurora from the OVATION Prime model) will be used in the assimilative procedure developed in this study. Three example distributions of Pedersen conductance from this model are shown in Figure 2, as contours plot over the high-latitude region in the Northern Hemisphere, plotted in geomagnetic latitude and MLT. In addition to the dayside enhancement, auroral enhancements with levels near 4–7 S are present. These auroral enhancements, which vary with varying solar wind conditions, are similar to others obtained throughout the 2 week time period of interest.

We should note that when conductance gradients are relatively small, as in our chosen model, the FAC is closely related to the vorticity of the plasma convection [e.g., Sofko *et al.*, 1995], and we can expect measurements related to the FAC (AMPERE) to give useful information about the convection, and measurements of the convection (SuperDARN) to give useful information about the FAC. This is demonstrated in the next section. However, undetermined conductance gradients degrade the accuracy of the information that AMPERE can provide about the convection and that SuperDARN can provide about the FAC. In particular, conductance gradients can shift the predicted locations of FAC relative to the convection. Quantification of these errors will require further study beyond the scope of this paper.

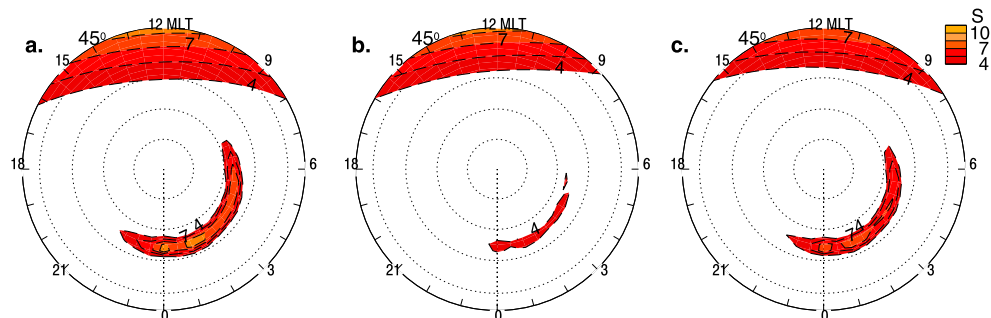


Figure 2. Example distributions of Pedersen conductance for (a) 1210 UT on 30 November 2011, (b) 0740 UT on 29 November 2011, and (c) 0050 UT on 29 November 2011, plotted in MLT and geomagnetic latitude poleward of 45°. Contour levels are given at 1.5 S intervals.

4. Mapping With SuperDARN and AMPERE

Using the conductance model described in section 3, the OI mapping procedure described in section 2.2 is applied to SuperDARN and AMPERE data. Figure 3 shows example results of the procedure for 1210–1220 UT on 30 November 2011 in the Northern Hemisphere. Distributions of electrostatic potential in kV, FAC density (positive downward) in $\mu\text{A}/\text{m}^2$, and Poynting flux in mW/m^2 are shown in Figures 3a and 3d, 3b and 3e, and 3c and 3f, respectively, where Poynting flux is defined as $S_{\parallel} = -\vec{E} \times \delta\vec{B}/\mu_0$, positive downward. The OI procedure is solved in terms of both electrostatic potential, with results shown in Figures 3a–3c, and magnetic potential, shown in Figures 3d–3f. Each panel in Figure 3 contains, in geomagnetic coordinates, a colored contour plot of the parameter of interest, with positive values in red and negative values in blue. In Figures 3a, 3b, 3d, and 3e, locations of AMPERE and SuperDARN observations used in the procedure are indicated by black dots, and the maximum and minimum values of the plotted parameter are given at the bottom right and left, respectively, of the map. In Figures 3c and 3f, the total Poynting flux integrated over the entire high-latitude region is given at the bottom right of the map, in GW. The orientation of the smoothed and time-shifted OMNI interplanetary magnetic field (IMF) for this time period is indicated by the red vector in the lower right portion in Figure 3. The Pedersen conductance distribution corresponding to this time period is shown in Figure 2a.

A two-cell convection pattern is seen in both Figures 3a and 3d, but more asymmetries between the dawn and dusk convection cells and a larger rotation of the pattern in MLT is seen in Figure 3a (from the electrostatic potential solution) than in Figure 3d (from the magnetic potential solution). A Region 1/Region 2 (R1/R2) configuration of FACs is seen in Figure 3e (from the magnetic potential solution), but not in Figure 3b (from the electrostatic potential solution), which contains smaller-scale, less well organized features as well as larger FAC densities in the polar cap region (poleward of $\sim 75^\circ$). The Poynting flux distribution also differs significantly between the two solutions, with the largest enhancements seen in the dayside cusp region for the electrostatic potential solution and in the dawn and dusk auroral zones for the magnetic potential solutions.

Due to the significant uncertainty in specifying the ionospheric conductance distribution, estimated distributions of electric field and electrostatic potential are more directly related to the electrostatic potential solution, which is more strongly constrained by the SuperDARN plasma drift observations, and estimated distributions of magnetic perturbations and FAC density are more directly related to the magnetic potential solution, which is more strongly constrained by the AMPERE magnetic perturbation observations (see Figure 1). This effect is expected to be especially important during increased geomagnetic activity, when the true conductance distribution can have large spatial and temporal variations that are not captured by the modeled conductance. These relations explain, in part, the differences between the results obtained from the electrostatic potential solution (Figures 3a–3c) and those obtained from the magnetic potential solution (Figures 3d–3f). Differences between the two sets of results are also caused by differences in the properties of the background model and background model error covariance used when solving in terms of magnetic potential versus electrostatic potential (as explained in section 2.2). In particular, the error covariance of the magnetic potential background model is characterized using fewer EOFs and a significantly smaller data set as compared to that for the electrostatic potential background model.

Because of the conductance-related uncertainties and the differences between the solutions for electrostatic potential and magnetic potential described above, the electrostatic potential distribution estimated using

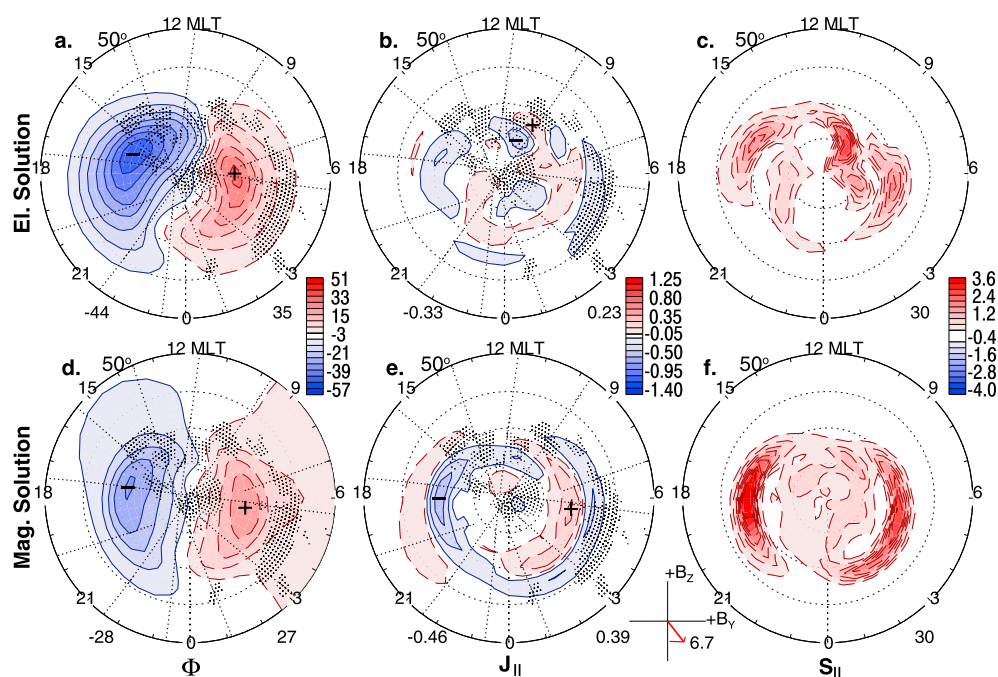


Figure 3. OI results for 1210–1220 UT on 30 November 2011 in the Northern Hemisphere. Distributions of (a and d) electrostatic potential in kV, (b and e) FAC density (positive downward) in $\mu\text{A}/\text{m}^2$, and (c and f) Poynting flux in mW/m^2 are shown as contour plots in geomagnetic coordinates, according to the color scales shown at right of each pair of maps. Results from both the (Figures 3a–3c) electrostatic potential and (Figures 3d–3f) magnetic potential solution are shown. Locations of AMPERE and SuperDARN observations are indicated by black dots. Maximum and minimum values of the plotted parameter are given at bottom right and left, respectively, of each map, except in Figures 3c and 3f, where the integrated Poynting flux in GW is given at bottom right of each map. The orientation of the prevailing IMF is indicated by the red vector in lower right, with the transverse magnitude given in nT.

the electrostatic potential solution (e.g., Figure 3a) is considered more reliable than that obtained from the magnetic potential solution (e.g., Figure 3f), and the FAC distribution estimated using the magnetic potential solution (e.g., Figure 3e) is considered more reliable than that obtained from the electrostatic potential solution (e.g., Figure 3b). Throughout the remainder of the study, therefore, the OI procedure solved in terms of electrostatic potential (based on both SuperDARN and AMPERE data) will be used to calculate electric field and electrostatic potential distributions, while the magnetic potential solution (based on both SuperDARN and AMPERE data) will be used to calculate magnetic perturbations and FAC distributions. The Poynting flux distribution will be calculated using electric fields from the electrostatic potential solution and magnetic perturbations from the magnetic potential solution. This approach signifies that the electric field, FAC, and Poynting flux distributions may not be completely physically self-consistent (due to the use of two different solutions), but the electric field and FAC distributions will each be more consistent with the observations than if one self-consistent solution was selected.

To demonstrate the benefit of incorporating both SuperDARN and AMPERE observations into the estimation of both electrostatic and magnetic potential, the results obtained as described above are compared to results obtained by simultaneously but independently estimating electrostatic potential from SuperDARN observations and estimating magnetic potential from AMPERE observations (similar to approaches used previously, e.g., Waters *et al.* [2004]). Figure 4 shows Northern Hemisphere distributions of electrostatic potential, FAC density, and Poynting flux, following the format in Figure 3, based on using the two data sets independently (Figures 4a–4c) and together (Figures 4d–4f) for estimating electrostatic and magnetic potential. The background color in Figures 4a, 4b, 4d, and 4e indicates estimated uncertainty in the solution in terms of electric field (in mV/m) for Figures 4a and 4d and in terms of magnetic perturbation (in nT) for Figures 4b and 4e, calculated using equation (5). A large gap in the SuperDARN data coverage is observed in the noon to dusk region in Figure 4a, and the duskside convection reversal is observed at very high latitudes, near the patch of available data in the polar cap. When AMPERE data are included in estimating the electrostatic potential distribution, the duskside convection reversal in the estimated convection pattern is more narrow and

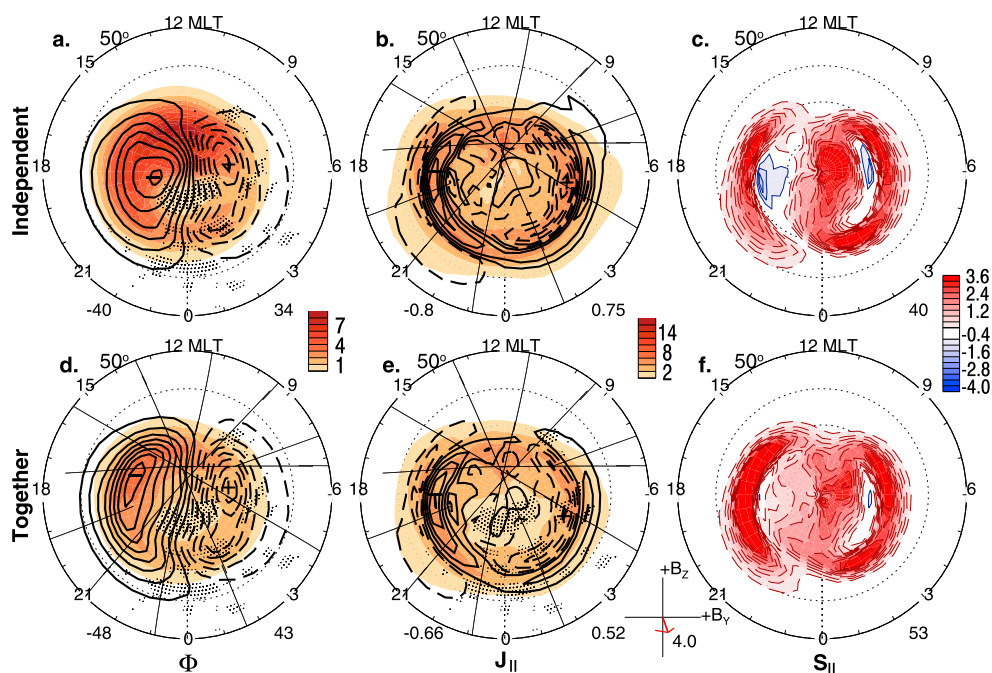


Figure 4. Distributions of electrostatic potential, FAC density, and Poynting flux for 0740–0750 UT, 29 November 2011, following the format in Figure 3. Background color in Figures 4a and 4b and in Figures 4d and 4e indicates estimated uncertainty in mV/m and nT, respectively, according to the color scales. Results are shown for using the SuperDARN and AMPERE data (a–c) independently and (d–f) together (see text for detail). The Pedersen conductance distribution corresponding to this time period is shown in Figure 2b.

elongated in MLT and is observed at more equatorward latitudes, and also, the estimated strength of convection is increased (as measured by the maximum and minimum values in the electrostatic potential distribution). The estimated uncertainty in the electric field values decreases significantly in the region lacking SuperDARN data when AMPERE data are included and fill this gap. The estimated FAC distribution does not change significantly in morphology when estimated using SuperDARN data in addition to AMPERE data, although the magnitudes decrease somewhat. The estimated uncertainties in the solution decrease, especially where SuperDARN data are present. The Poynting flux distribution is similar in morphology between the two estimates, but the regions of negative (upward) Poynting flux observed in the estimate obtained from the two data sets used independently largely disappears and the magnitudes are higher in the dawn and dusk auroral regions when the data sets are used together, such that the total integrated Poynting flux increases by over 30%. These changes in the Poynting flux distribution likely result in part from the closer correspondence between the locations of the convection reversal and the large-scale R1/R2 FACs when both data sets are incorporated into estimating both electrostatic and magnetic potential. It should be noted, however, that the reduction in negative Poynting flux could be, in part, an artifact of the conductance distribution that is used when incorporating both data sets. In particular, the background nightside conductance level is significantly increased such that the spatial distribution is more uniform; reduced gradients in the conductance distribution result in more correlated electric field and magnetic perturbation distributions and thus more positive Poynting flux.

Another example of using AMPERE and SuperDARN data together versus independently is shown in Figure 5, following the format in Figure 4. In this example, there is a large gap in SuperDARN data coverage from dawn to noon, and there is also a gap in AMPERE data in the duskside auroral region. Here the electrostatic potential distribution does not change significantly when AMPERE data are included in the estimate, although the estimated uncertainty in the solution is reduced in the region without SuperDARN data. The estimated FAC distribution, however, does change when SuperDARN data are included; small-scale FAC structures that appeared in the region missing AMPERE data are significantly attenuated when SuperDARN data fill in the AMPERE data gap, and the estimated uncertainty in the solution is significantly reduced. The estimated

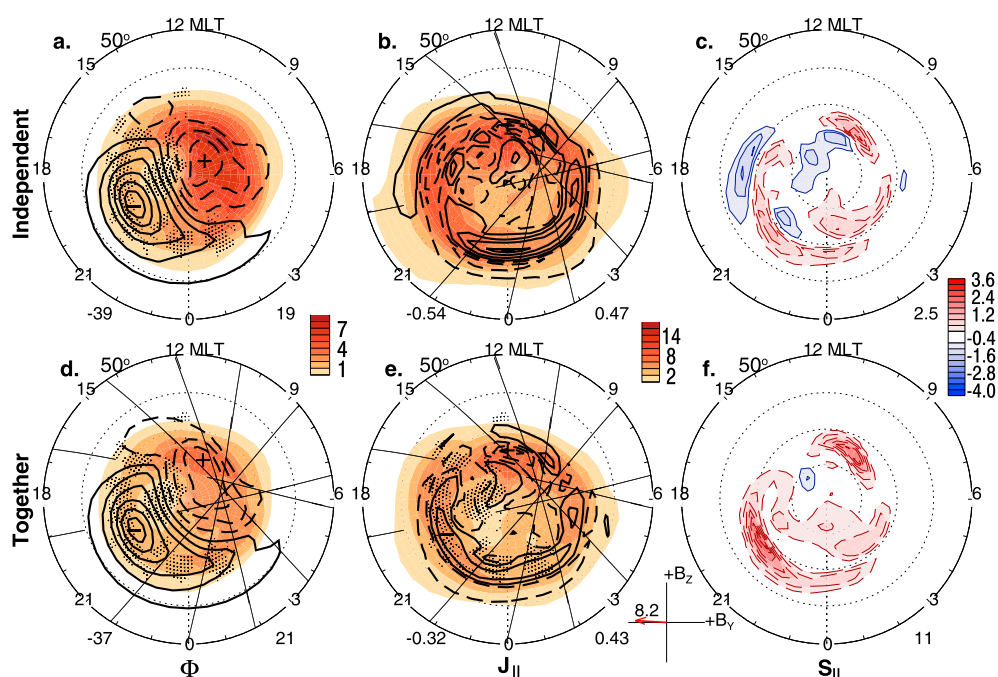


Figure 5. Distributions of electrostatic potential and FAC density, with uncertainties, and of Poynting flux for 0050–0100 UT, 29 November 2011, following the format in Figure 4. The Pedersen conductance distribution corresponding to this time period is shown in Figure 2c.

Poynting flux distribution is significantly different between the two estimates, with fewer regions of negative flux observed and enhanced magnitudes in positive flux regions when the data sets are used together.

Impacts of using both SuperDARN and AMPERE data to estimate both electrostatic and magnetic potential such as shown in Figures 4 and 5 are seen in both hemispheres throughout the 1 week study period. Because generally less SuperDARN data are available in the Southern Hemisphere than in the Northern Hemisphere, the impact of including AMPERE data in the electrostatic potential estimate is greater and the impact of including SuperDARN data in the magnetic potential estimate is lesser in the South than in the North. Comparing results of using both data sets together versus independently, the median estimated uncertainty in the electrostatic and magnetic potential solutions decrease by 8% and 37%, respectively, in the Northern Hemisphere and by 15% and 22%, respectively, in the Southern Hemisphere. The median cross-polar cap potential drop (CPCP) increases by 4% and 34% in the Northern and Southern Hemispheres, respectively, when both data sets are used together, while the total integrated upward FAC density increases in the Northern Hemisphere by 11% and decreases in the Southern Hemisphere by 7%. The total integrated Poynting flux increases by 33% and 47% in the Northern and Southern Hemispheres, respectively, when both data sets are used together versus independently. Similar impacts can be expected for other times with weak to moderate geomagnetic disturbances and moderate to good SuperDARN data coverage (as in the study period).

5. Example Results

To demonstrate the OI mapping procedure developed in this study, it is applied using both SuperDARN and AMPERE data from both hemispheres for every 10 min time step in the whole week-long study period. Time series of key parameters derived from the results for the middle 3 days (28–30 November 2011) are shown in Figure 6, along with the number of data points available and the 45 min smoothed (but not time shifted) OMNI solar wind data. Vertical dotted lines indicate times considered later in this section. During these days, the CPCP, the total integrated upward FAC density, and the total integrated Poynting flux are all larger in the Southern (summer) Hemisphere than in the Northern (winter) Hemisphere. The larger FAC density in the summer hemisphere is consistent with previous studies using satellite magnetometer data [e.g., Weimer, 2001; Christiansen et al., 2002; Papitashvili et al., 2002; Ohtani et al., 2005; Green et al., 2009] and is related to the larger polar conductances in summer than winter. That the estimated CPCP is also usually larger in the Southern (summer) Hemisphere in Figure 6 contrasts with the Northern Hemisphere observations of

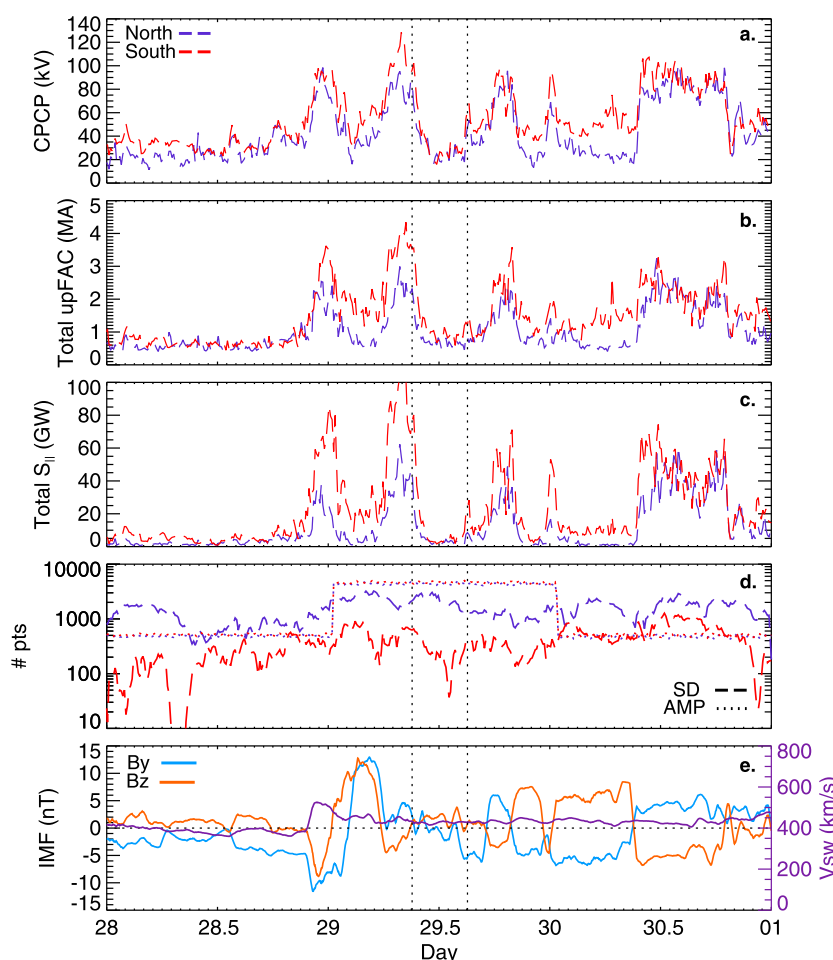


Figure 6. Time series of (a) CPCP, (b) integrated upward FAC density, (c) integrated Poynting flux, (d) number of SuperDARN and AMPERE data points, and (e) OMNI solar wind parameters for 28–30 November 2011. In Figures 6a–6d, values are shown for both the (blue) Northern and (red) Southern Hemispheres.

de la Beaujardière et al. [1991], who found somewhat larger winter than summer CPCP, but is consistent with the results of *Pettigrew et al.* [2010], who found that the estimated CPCP (using SuperDARN data) is usually larger in the summer hemisphere than in the winter hemisphere. The generally larger Southern summer than Northern winter CPCP in Figure 6 may be due partly to geophysical effects but is probably also influenced by hemispheric asymmetry of the data coverage: more SuperDARN data in the Northern Hemisphere and densest AMPERE coverage near the geographic pole, which in the Southern Hemisphere lies at a lower magnetic latitude, where FAC tends to be larger than it is inside the polar cap. AMPERE data play a greater relative role, compared with SuperDARN data, in the estimation of the Southern CPCP, and the influence of AMPERE data on the CPCP estimation depends much more sensitively on the estimated conductance than does the influence of SuperDARN data. Thus, there can be hemispheric biases in the CPCP estimation. Nevertheless, strong inter-hemispheric differences seen in Figure 6 are likely representative of real geophysical effects, at least in part. The largest relative interhemispheric differences in the CPCP, integrated upward FAC, and integrated Poynting flux are observed under northward IMF conditions with a strong IMF B_y component (see, for example, the first half of 30 November). This is consistent with the concept that stronger electric potential differences and FAC can be produced in the hemisphere tilted toward the Sun when the IMF is strongly northward [e.g., *Crooker and Rich*, 1993; *Pettigrew et al.*, 2010]. The largest relative interhemispheric differences between these parameters are observed under northward IMF conditions with a strong IMF B_y component (see, for example, the first half of 30 November). Clear and often abrupt changes are seen in these parameters in response to changes in the solar wind conditions; these responses are not examined in this paper but are shown to demonstrate possible applications of the technique developed in this study.

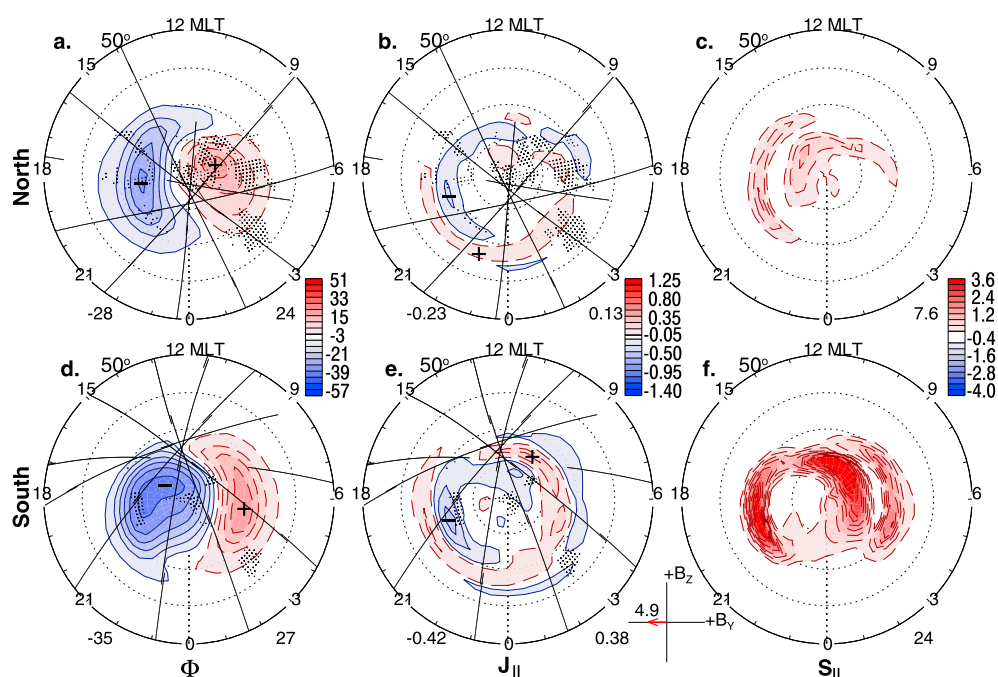


Figure 7. Distributions of electrostatic potential, FAC density, and Poynting flux for 1500–1510 UT, 29 November 2011, following the format in Figure 3. Results are shown for (a–c) the Northern Hemisphere and (d–f) the Southern Hemisphere.

Results from both hemispheres from an example snapshot (29 November, 1500 UT) are shown in Figure 7, following the format in Figure 3. The IMF is B_y dominated at this time, and both the convection pattern and the dayside FAC distribution have different morphologies in the two hemispheres, as expected [e.g., *Feldstein et al.*, 1984; *Heppner and Maynard*, 1987]. The electrostatic potential and FAC density values are larger in the South than in the North, and the Poynting flux is significantly enhanced in the South, especially in the polar

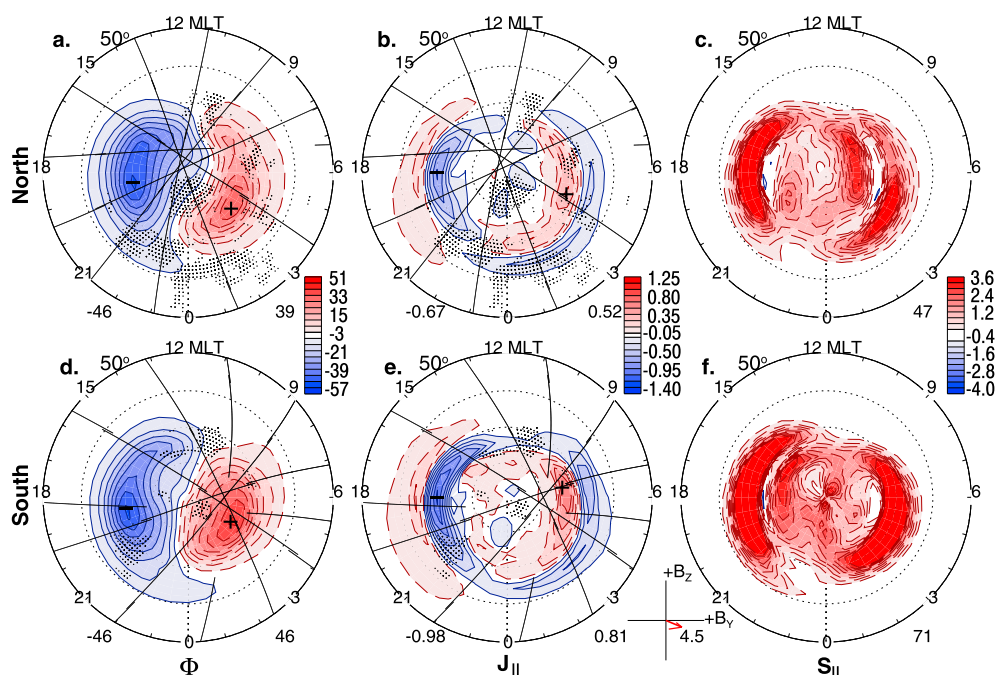


Figure 8. Distributions of electrostatic potential, FAC density, and Poynting flux for 0900–0910 UT, 29 November 2011, following the format in Figure 7.

cap region. The interhemispheric differences in magnitudes of electrostatic potential and FACs are consistent with winter-like/summer-like differences observed previously with SuperDARN data [Pettigrew *et al.*, 2010] and satellite magnetic perturbation data [e.g., Papitashvili *et al.*, 2002].

Another example snapshot is shown in Figure 8 for 29 November, 0900 UT, in the format in Figure 7. Although the IMF is weak and IMF B_z is near zero, an enhanced two-cell convection pattern with moderate R1 and R2 currents is observed, similar to that expected for southward IMF conditions. A sharp convection reversal is observed near midnight just poleward of 60° in both hemispheres, as indicated by the narrow extension of the negative (blue) potential cell toward midnight in the Northern Hemisphere and across midnight in the Southern Hemisphere. These features, which are well-defined by the combined SuperDARN and AMPERE data sets, would not have been predicted by an empirical model keyed to the solar wind conditions nor by the SuperDARN data alone (note in particular the absence of SuperDARN data near midnight in the Southern Hemisphere).

6. Summary

In this study, an OI data assimilation procedure is developed for obtaining complete maps of electrodynamic parameters in the high-latitude ionosphere based on AMPERE observations of magnetic perturbations and SuperDARN observations of plasma drifts. This procedure incorporates realistic background model error properties, estimated using EOF analysis of both SuperDARN and AMPERE observations. In this study, unlike previous assimilative mapping procedures, solutions are obtained in terms of both electrostatic potential and vector magnetic potential. Using magnetic potential in addition to electrostatic potential is shown to enable an improved specification of magnetic perturbation and FAC distributions.

Because knowledge of the ionospheric conductance is required for the mapping procedure, various techniques for estimating and specifying conductance are implemented and tested. Estimating or adjusting conductance based on the AMPERE and SuperDARN observations was found to give poor results. Based on the results of quantitative tests, the selected method of specifying conductance includes an inflated night-side conductance level of 4 S and diffuse aurora from the OVATION Prime empirical model [Newell *et al.*, 2009, 2010], in addition to a parametrized model of solar-produced conductance. This element of the mapping procedure contains a large degree of uncertainty, and future work to improve the mapping procedure could incorporate conductance-related observations to improve its specification.

Several benefits of combining SuperDARN and AMPERE data are demonstrated, including improved specification of features in regions lacking one type of data (SuperDARN or AMPERE) and better correspondence of features in the electrostatic potential distribution to those in the FAC distribution. Additionally, the estimated uncertainty in the mapped results decreases by $\sim 20\%$ when the two data sets are used together rather than independently.

The SuperDARN and AMPERE data sets have great utility due to their routine availability at relatively high time resolution and large spatial coverage, and the mapping procedure developed in this study increases their utility by optimally combining the two data sets to obtain complete and consistent distributions of electrodynamic parameters of interest. As briefly demonstrated, the mapping procedure can be used, among other applications, for investigating solar wind-magnetosphere-ionosphere coupling processes and for estimating the energy and momentum input to the upper atmosphere in the high-latitude region.

References

- Aksnes, A., J. Stadsnes, J. Bjørndal, N. Østgaard, R. R. Vondrak, D. L. Detrick, T. J. Rosenberg, G. A. Germany, and D. Chenette (2002), Instantaneous ionospheric global conductance maps during an isolated substorm, *Ann. Geophys.*, **20**, 1181–1191.
- Anderson, B. J., H. Korth, C. L. Waters, D. L. Green, V. G. Merkin, R. J. Barnes, and L. P. Dyrd (2014), Development of large-scale Birkeland currents determined from the Active Magnetosphere and Planetary Electrodynamics Response Experiment (AMPERE), *Geophys. Res. Lett.*, **41**, 3017–3025, doi:10.1002/2014GL059941.
- Baker, K. B., and S. Wing (1989), A new magnetic coordinate system for conjugate studies at high latitudes, *J. Geophys. Res.*, **94**, 9139–9143.
- Blomberg, L. G., and G. T. Marklund (1988), The influence of conductivities consistent with field-aligned currents on high-latitude convection patterns, *J. Geophys. Res.*, **93**, 14,493–14,499.
- Brekke, A., and J. Moen (1993), Observations of high latitude ionospheric conductances, *J. Atmos. Terr. Phys.*, **55**, 1493–1512, doi:10.1016/0021-9169(93)90126-J.
- Chisham, G., et al. (2007), A decade of the Super Dual Auroral Radar Network (SuperDARN): Scientific achievements, new techniques and future directions, *Surv. Geophys.*, **28**, 33–109.
- Chisham, G., T. K. Yeoman, and G. J. Sofko (2008), Mapping ionospheric backscatter measured by the SuperDARN HF radars—Part 1. A new empirical virtual height model, *Ann. Geophys.*, **26**, 823–841.

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- Christiansen, F., V. O. Papitashvili, and T. Neubert (2002), Seasonal variations of high-latitude field-aligned currents inferred from Ørsted and Magsat observations, *J. Geophys. Res.*, **107**, 1029, doi:10.1029/2001JA00104.
- Coumans, V., J.-C. Gérard, B. Hubert, S. B. Mende, and S. W. H. Cowley (2004), Morphology and seasonal variations of global auroral proton precipitation observed by IMAGE-FUV, *J. Geophys. Res.*, **109**, A12205, doi:10.1029/2003JA010348.
- Cousins, E. D. P., and S. G. Shepherd (2010), A dynamical model of high-latitude convection derived from SuperDARN plasma drift measurements, *J. Geophys. Res.*, **115**, A12329, doi:10.1029/2010JA016017.
- Cousins, E. D. P., T. Matsuo, and A. D. Richmond (2013a), SuperDARN assimilative mapping, *J. Geophys. Res. Space Physics*, **118**, 7954–7962, doi:10.1002/2013JA019321.
- Cousins, E. D. P., T. Matsuo, and A. D. Richmond (2013b), Mesoscale and large-scale variability in high-latitude ionospheric convection: Dominant modes and spatial/temporal coherence, *J. Geophys. Res. Space Physics*, **118**, 7895–7904, doi:10.1002/2013JA019319.
- Cousins, E. D. P., T. Matsuo, A. D. Richmond, and B. J. Anderson (2015), Dominant modes of variability in large-scale Birkeland currents, *J. Geophys. Res. Space Physics*, **120**, doi:10.1002/2014JA020462.
- Crooker, N. U., and F. J. Rich (1993), Lobe cell convection as a summer phenomenon, *J. Geophys. Res.*, **98**, 13,403–13,407.
- Davies, J. A., and M. Lester (1999), The relationship between electric fields, conductances and currents in the high-latitude ionosphere: A statistical study using EISCAT data, *Ann. Geophys.*, **17**, 43–52.
- de la Beaujardière, O., D. Alcaydé, J. Fontanari, and C. Leger (1991), Seasonal dependence of high-latitude electric fields, *J. Geophys. Res.*, **96**(A4), 5723–5735.
- Drayton, R. A., A. V. Koustov, M. R. Hairston, and J. -P. Villain (2005), Comparison of DMSP cross-track ion drifts and SuperDARN line-of-sight velocities, *Ann. Geophys.*, **23**, 2479–2486.
- Feldstein, Y. I., R. G. Afonina, B. A. Belov, A. E. Levitin, D. S. Faermark, and V. Y. Gaidukov (1984), High-latitude field-aligned currents patterns in connection with magnetospheric structure, in *Magnetospheric Currents*, *Geophys. Monogr. Ser.*, vol. 28, edited by T. A. Potemra, pp. 284–293, AGU, Washington, D. C.
- Fuller-Rowell, T. J., and D. S. Evans (1987), Height-integrated Pedersen and Hall conductivity patterns inferred from the TIROS-NOAA satellite data, *J. Geophys. Res.*, **92**, 7606–7618.
- Germany, G. A., D. G. Torr, P. G. Richards, M. R. Torr, and S. Johns (1994), Determination of ionospheric conductivities from FUV auroral emissions, *J. Geophys. Res.*, **99**, 23,297–23,305.
- Germany, G. A., D. Lummerzheim, and P. G. Richards (2001), Impact of model differences on quantitative analysis of FUV auroral emissions: Total ionization cross sections, *J. Geophys. Res.*, **106**, 12,837–12,843.
- Gillies, R. G., G. C. Hussey, G. J. Sofko, K. A. McWilliams, R. A. D. Fiori, P. Ponomarenko, and J.-P. St.-Maurice (2009), Improvement of SuperDARN velocity measurements by estimating the index of refraction in the scattering region using interferometry, *J. Geophys. Res.*, **114**, A07305, doi:10.1029/2008JA013967.
- Gillies, R. G., G. C. Hussey, G. J. Sofko, P. Ponomarenko, and K. A. McWilliams (2011), Improvement of HF coherent radar line-of-sight velocities by estimating the refractive index in the scattering volume using radar frequency shifting, *J. Geophys. Res.*, **116**, A01302, doi:10.1029/2010JA016043.
- Green, D. L., C. L. Waters, H. Korth, B. J. Anderson, A. J. Ridley, and R. J. Barnes (2007), Technique: Large-scale ionospheric conductance estimated from combined satellite and ground-based electromagnetic data, *J. Geophys. Res.*, **112**, A05303, doi:10.1029/2006JA012069.
- Green, D. L., C. L. Waters, B. J. Anderson, and H. Korth (2009), Seasonal and interplanetary magnetic field dependence of the field-aligned currents for both Northern and Southern Hemispheres, *Ann. Geophys.*, **27**, 1701–1715.
- Greenwald, R. A., et al. (1995), DARN/SuperDARN: A global view of the dynamics of high-latitude convection, *Space Sci. Rev.*, **71**, 761–796.
- Hardy, D. A., M. S. Gussenhoven, R. Raistrick, and W. J. McNeil (1987), Statistical and functional representations of the pattern of auroral energy flux, number flux, and conductivity, *J. Geophys. Res.*, **92**, 12,275–12,294.
- Heppner, J. P., and N. C. Maynard (1987), Empirical high-latitude electric field models, *J. Geophys. Res.*, **92**, 4467–4489.
- Kalnay, E. (2003), *Atmospheric Modeling, Data Assimilation and Predictability*, Cambridge Univ. Press, Cambridge, U. K.
- Knipp, D. J., T. Matsuo, L. Kilcommons, A. Richmond, B. Anderson, H. Korth, R. Redmon, B. Mero, and N. Parrish (2014), Comparison of magnetic perturbation data from LEO satellite constellations: Statistics of DMSP and AMPERE, *Space Weather*, **12**, 2–23, doi:10.1002/2013SW000987.
- Langel, R. A. (1974), Near-Earth magnetic disturbance in total field at high latitudes: 1. Summary of data from Ogo 2, 4, and 6, *J. Geophys. Res.*, **79**, 2363–2371.
- Lester, M. (2013), The Super Dual Auroral Radar Network (SuperDARN): An overview of its development and science, *Adv. Polar Sci.*, **24**, 1–11.
- Lummerzheim, D., and J. Lilén (1994), Electron transport and energy degradation in the ionosphere: Evaluation of the numerical solution, comparison with laboratory experiments and auroral observations, *Ann. Geophys.*, **12**, 1039–1051.
- Lummerzheim, D., M. H. Rees, J. D. Craven, and L. A. Frank (1991), Ionospheric conductances derived from DE-1 auroral images, *J. Atmos. Terr. Phys.*, **53**, 281–292.
- Marklund, G., L. G. Blomberg, K. Stasiewicz, J. S. Murphree, R. Potelette, L. J. Zanetti, T. A. Potemra, D. A. Hardy, and F. J. Rich (1988), Snapshots of high-latitude electrodynamics using Viking and DMSP F7 observations, *J. Geophys. Res.*, **93**, 14,479–14,492.
- Matsuo, T., A. D. Richmond, and G. Lu (2005), Optimal interpolation analysis of high-latitude ionospheric electrodynamics using empirical orthogonal functions: Estimation of dominant modes of variability and temporal scales of large-scale electric fields, *J. Geophys. Res.*, **110**, A06301, doi:10.1029/2004JA010531.
- Merkin, V. G., G. Milikh, K. Papadopoulos, J. Lyon, Y. S. Dimant, A. S. Sharma, C. Goodrich, and M. Wiltberger (2005), Effect of anomalous electron heating on the transpolar potential in the LFM global MHD model, *Geophys. Res. Lett.*, **32**, L22101, doi:10.1029/2005GL023315.
- Mitchell, E. J., P. T. Newell, J. W. Gjerloev, and K. Liou (2013), OVATION-SM: A model of auroral precipitation based on SuperMAG generalized auroral electrojet and substorm onset times, *J. Geophys. Res. Space Physics*, **118**, 3747–3759, doi:10.1002/jgra.50343.
- Newell, P. T., T. Sotirelis, and S. Wing (2009), Diffuse, monoenergetic, and broadband aurora: The global precipitation budget, *J. Geophys. Res.*, **114**, A09207, doi:10.1029/2009JA014326.
- Newell, P. T., T. Sotirelis, and S. Wing (2010), Seasonal variations in diffuse, monoenergetic, and broadband aurora, *J. Geophys. Res.*, **115**, A03216, doi:10.1029/2009JA014805.
- Ohtani, S., G. Ueno, and T. Higuchi (2005), Comparison of large-scale field-aligned currents under sunlit and dark ionospheric conditions, *J. Geophys. Res.*, **110**, A09230, doi:10.1029/2005JA011057.
- Papitashvili, V. O., F. Christiansen, and T. Neubert (2002), A new model of field-aligned currents derived from high-precision satellite magnetic field data, *Geophys. Res. Lett.*, **29**, 1683, doi:10.1029/2001GL014207.
- Pettigrew, E. D., S. G. Shepherd, and J. M. Ruohoniemi (2010), Climatological patterns of high-latitude convection in the Northern and Southern Hemispheres: Dipole tilt dependencies and interhemispheric comparisons, *J. Geophys. Res.*, **115**, A07305, doi:10.1029/2009JA014956.

- Richmond, A. D. (1995a), Ionospheric electrodynamics using magnetic Apex coordinates, *J. Geomagn. Geoelectr.*, *47*, 191–212.
- Richmond, A. D. (1995b), Ionospheric electrodynamics, in *Handbook of Atmospheric Electrodynamics*, vol. 2, edited by H. Volland, pp. 249–290, CRC Press, Boca Raton, Fla.
- Richmond, A. D., and Y. Kamide (1988), Mapping electrodynamic features of the high-latitude ionosphere from localized observations: Technique, *J. Geophys. Res.*, *93*, 5741–5759.
- Ridley, A. J., T. I. Gombosi, and D. L. DeZeeuw (2004), Ionospheric control of the magnetosphere: Conductance, *Ann. Geophys.*, *22*, 567–584.
- Robinson, R. M., R. R. Vondrak, K. Miller, T. Dabbs, and D. Hardy (1987), On calculating ionospheric conductances from the flux and energy of precipitating electrons, *J. Geophys. Res.*, *92*, 2565–2569.
- Ruohoniemi, J. M., and K. B. Baker (1998), Large-scale imaging of high-latitude convection with Super Dual Auroral Radar Network HF radar observations, *J. Geophys. Res.*, *103*, 20,797–20,811.
- Ruohoniemi, J. M., and R. A. Greenwald (1997), Rates of scattering occurrence in routine HF radar observations during solar cycle maximum, *Radio Sci.*, *32*, 1051–1070.
- Sofko, G. J., R. Greenwald, and W. Bristow (1995), Direct determination of large-scale magnetospheric field-aligned currents with SuperDARN, *Geophys. Res. Lett.*, *22*, 2041–2044.
- Vennerstrom, S., T. Moretto, L. Rastätter, and J. Raeder (2006), Modeling and analysis of solar wind generated contributions to the near-Earth magnetic field, *Earth Planets Space*, *58*, 451–461.
- Wallis, D. D., and E. E. Budzinski (1981), Empirical models of height integrated conductivities, *J. Geophys. Res.*, *86*, 125–137.
- Waters, C. L., B. J. Anderson, R. A. Greenwald, R. J. Barnes, and J. M. Ruohoniemi (2004), High-latitude poynting flux from combined Iridium and SuperDARN data, *Ann. Geophys.*, *22*, 2861–2875.
- Weimer, D. R. (2001), An improved model of ionospheric electric potentials including perturbations and application to the Geospace Environment Modeling November 24, 1996, event, *J. Geophys. Res.*, *106*, 407–416.